Chapter 5
Dynamics of Many-particle System

In this chapter we will extend application of dynamic laws from a single particle to a system of particles (质点系).

- **the tendency of motion of the system rather than individual pieces — particles.**
  Concept of center of mass
- **the system with variable mass — open system**
- **collision**
- **fluid motion**
- **symmetry and conservation***

5.1 Center of mass and C-frame

$$m_1 \ddot{\boldsymbol{r}}_1 = \boldsymbol{F}_1 + \boldsymbol{F}_{12} + \boldsymbol{F}_{13} + \cdots + \boldsymbol{F}_{1N}$$

or

$$m_i \ddot{\boldsymbol{r}}_i = \boldsymbol{F}_i + \sum_{\substack{j=1 \\ j \neq i}}^{N} \boldsymbol{F}_{ij}$$

$$\sum_i$$

$$\sum_i m_i \ddot{\boldsymbol{r}}_i = \sum_i \boldsymbol{F}_i + \sum_{\substack{i,j \\ i \neq j}} \boldsymbol{F}_{ij}$$

$$\sum_i \boldsymbol{F}_i \equiv \boldsymbol{F}_{\text{ext}}$$

All paired

Define

center of mass (COM)

$$\sum_i m_i \, \boldsymbol{r}_i \equiv \left(\sum_i m_i \right) \boldsymbol{r}_\mathrm{C}$$

$$= m_\mathrm{C} \, \boldsymbol{r}_\mathrm{C}$$

acting on system as a whole

$$\boxed{m_\mathrm{C} \, \ddot{\boldsymbol{r}}_\mathrm{C} = \boldsymbol{F}_\mathrm{ext}} \qquad \sum_{\substack{j=1 \\ j \neq i}}^{N} \boldsymbol{F}_{i\,j} = 0$$

Total momentum

$$\boldsymbol{p} = \sum \boldsymbol{p}_i = \sum m_i \, \dot{\boldsymbol{r}}_i = \frac{\mathrm{d}}{\mathrm{d}t} \sum m_i \, \boldsymbol{r}_i = m_\mathrm{C} \, \dot{\boldsymbol{r}}_\mathrm{C}$$

$$\boxed{\dot{\boldsymbol{p}} = \boldsymbol{F}_\mathrm{ext}}$$

The center of mass

$$\boldsymbol{r}_C = \frac{1}{m_C}\sum_i m_i \boldsymbol{r}_i$$

$$= \frac{\sum_i m_i \boldsymbol{r}_i}{\sum_i m_i}$$

$$= \sum_i \frac{m_i}{m_C}\boldsymbol{r}_i$$

weighted average

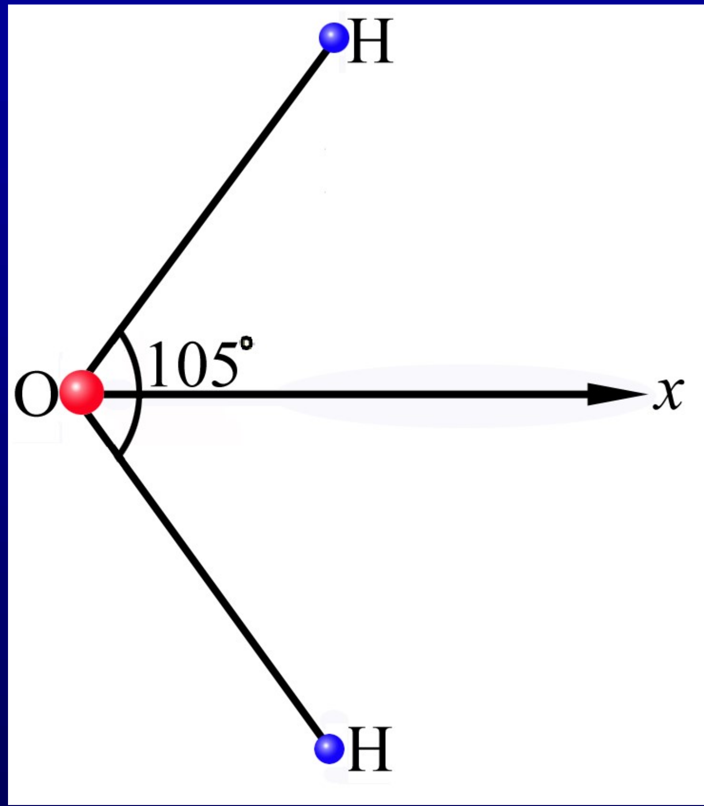
加权平均

Example 5.1 water molecule

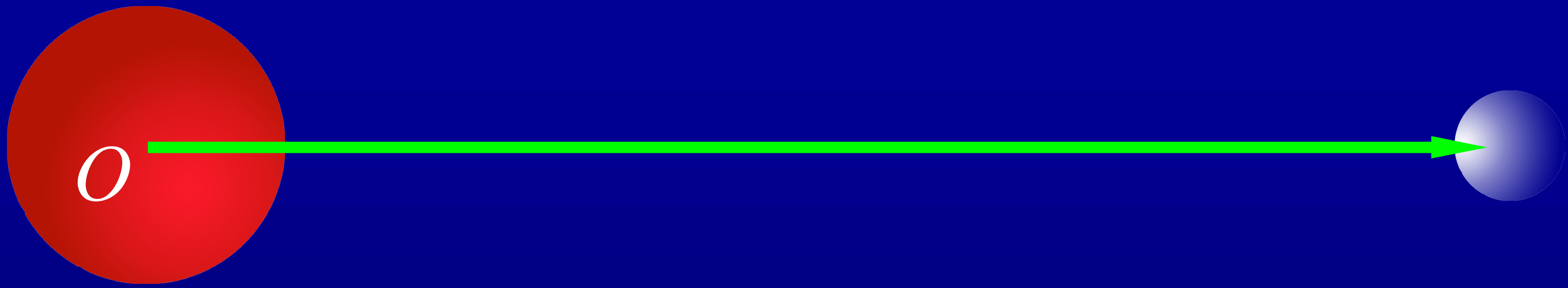
$$d = 0.91 \times 10^{-10}\,\mathrm{m}$$

$$x_C = \frac{2m_H \; d\cos\dfrac{105^\circ}{2}}{2m_H + m_O}$$

$$= 0.62 \times 10^{-11}\,\mathrm{m}$$



Example 5.2 earth-sun system



$$x_{\mathrm{C}} = \frac{m_{\oplus}}{m_{\odot} + m_{\oplus}} R_{\mathrm{ES}}$$

$$= \frac{5.98 \times 10^{24}\,\mathrm{kg}}{1.99 \times 10^{30}\,\mathrm{kg} + 5.98 \times 10^{24}\,\mathrm{kg}} 1.50 \times 10^{11}\,\mathrm{km}$$

$$= 4.49 \times 10^{5}\,\mathrm{km}$$

$$= 0.645 \times 10^{-3}\,R_{\odot}$$

For continuous distribution of mass

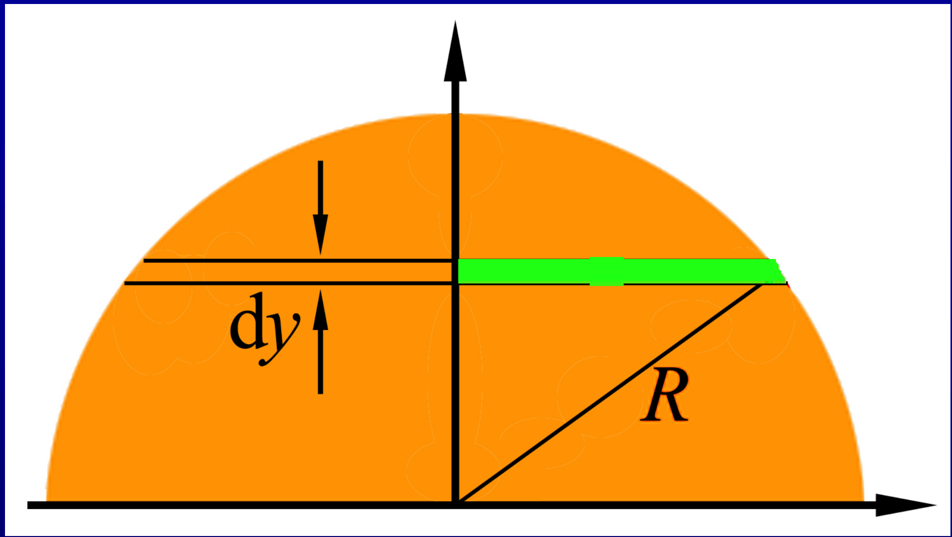
$$\boldsymbol{r}_{\mathrm{C}} = \frac{1}{m_{\mathrm{C}}} \sum_{i} m_i \boldsymbol{r}_i$$

$$\boldsymbol{r}_{\mathrm{C}} = \frac{1}{m_{\mathrm{C}}} \int \boldsymbol{r} \, \mathrm{d}m$$

$$= \frac{1}{m_{\mathrm{C}}} \int \boldsymbol{r} \rho \, \mathrm{d}\tau$$

mainly used for solid bodies

Example



$$x_C = 0$$

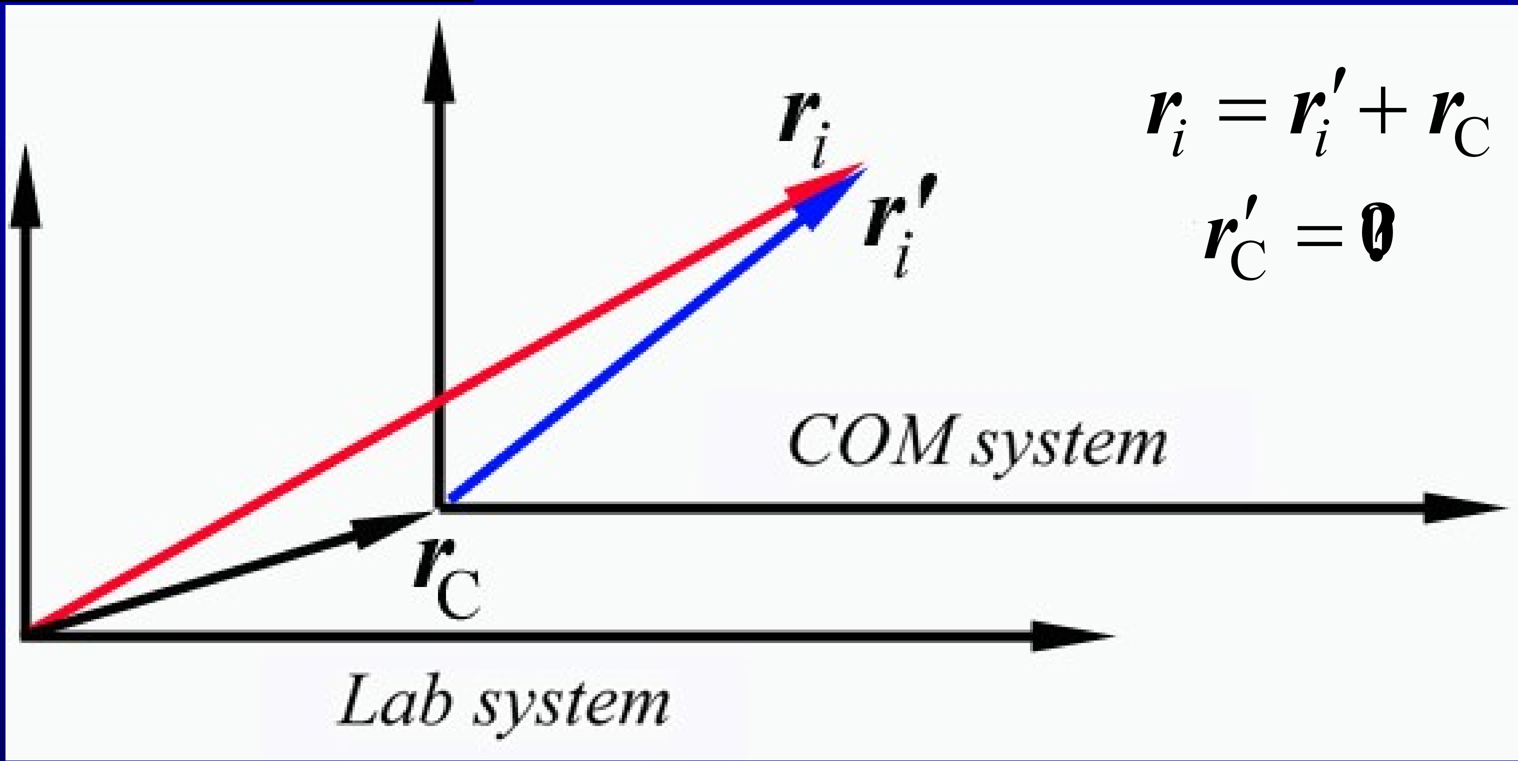
$$y_C = ?$$

$$\rho = \frac{m}{\frac{1}{2}\pi R^2}$$

$$\int \mathrm{d}\tau \to \int \mathrm{d}S = \int \mathrm{d}y \int \mathrm{d}x$$

$$= \int \mathrm{d}y \times 2\sqrt{R^2 - y^2}$$

COM system



$$\boldsymbol{p}' = \sum m_i \dot{\boldsymbol{r}}_i' = \frac{\mathrm{d}}{\mathrm{d}t}\left(\sum m_i \boldsymbol{r}_i'\right) = \frac{\mathrm{d}}{\mathrm{d}t}\left(m_C \boldsymbol{r}_C'\right) = 0$$

It is main characteristic of COM system.

$$\boldsymbol{L} = \sum_i \boldsymbol{r}_i \times \boldsymbol{p}_i$$

$$\frac{\mathrm{d}}{\mathrm{d}t}\boldsymbol{L} = \sum_i \boldsymbol{r}_i \times \left(\boldsymbol{F}_i + \sum_{\substack{j \\ j \neq i}} \boldsymbol{F}_{i\,j} \right)$$

$$\boxed{\frac{\mathrm{d}}{\mathrm{d}t}\boldsymbol{L} = \boldsymbol{M}_{\mathrm{ext}}}$$

$$\sum_{\substack{i,j \\ i \neq j}} \boldsymbol{r}_i \times \boldsymbol{F}_{i\,j} = \sum_{\substack{j,i \\ j \neq i}} \boldsymbol{r}_j \times \boldsymbol{F}_{j\,i} = -\sum_{\substack{i,j \\ i \neq j}} \boldsymbol{r}_j \times \boldsymbol{F}_{i\,j}$$

$$= \frac{1}{2} \sum_{\substack{i,j \\ i \neq j}} \left(\boldsymbol{r}_i - \boldsymbol{r}_j \right) \times \boldsymbol{F}_{ij}$$

$$= \mathbf{0}$$

Newton's 3rd law in strong form

* $$\boxed{\frac{\mathrm{d}}{\mathrm{d}t}\boldsymbol{L}_{\mathrm{C}} = \boldsymbol{M}_{\mathrm{C}}}$$

$$\boldsymbol{L} = \sum_i \boldsymbol{r}_i \times \boldsymbol{p}_i = \sum_i \left(\boldsymbol{r}_i' + \boldsymbol{r}_C\right) \times \boldsymbol{p}_i = \sum_i \boldsymbol{r}_i' \times \boldsymbol{p}_i + \boldsymbol{r}_C \times \sum_i \boldsymbol{p}_i$$

$$= \sum_i \boldsymbol{r}_i' \times m_i \left(\dot{\boldsymbol{r}}_i' + \dot{\boldsymbol{r}}_C\right) + \boldsymbol{r}_C \times \boldsymbol{p} = \sum_i \boldsymbol{r}_i' \times m_i \dot{\boldsymbol{r}}_i' + \left(\sum_i m_i \boldsymbol{r}_i'\right) \times \dot{\boldsymbol{r}}_C + \boldsymbol{r}_C \times \boldsymbol{p}$$

$$= \boldsymbol{L}_C + m_C \boldsymbol{r}_C' \times \dot{\boldsymbol{r}}_C + \boldsymbol{r}_C \times \boldsymbol{p} = \boldsymbol{L}_C + \boldsymbol{r}_C \times \boldsymbol{p}, \qquad (\boldsymbol{r}_C' = 0)$$

$$\boldsymbol{M}_{\text{ext}} = \sum_i \boldsymbol{r}_i \times \boldsymbol{F}_i = \sum_i \left(\boldsymbol{r}_i' + \boldsymbol{r}_C\right) \times \boldsymbol{F}_i = \sum_i \boldsymbol{r}_i' \times \boldsymbol{F}_i + \boldsymbol{r}_C \times \sum_i \boldsymbol{F}_i$$

$$= \boldsymbol{M}_C + \boldsymbol{r}_C \times \boldsymbol{F}_{\text{ext}},$$

$$\frac{d\boldsymbol{L}}{dt} = \frac{d\boldsymbol{L}_C}{dt} + \dot{\boldsymbol{r}}_C \times \boldsymbol{p} + \boldsymbol{r}_C \times \frac{d\boldsymbol{p}}{dt} = \frac{d\boldsymbol{L}_C}{dt} + \boldsymbol{r}_C \times \frac{d\boldsymbol{p}}{dt}$$

$$= \boldsymbol{M}_{\text{ext}} = \boldsymbol{M}_C + \boldsymbol{r}_C \times \boldsymbol{F}_{\text{ext}},$$

$$\because \quad \frac{d\boldsymbol{p}}{dt} = \boldsymbol{F}_{\text{ext}}, \quad \therefore \frac{d\boldsymbol{L}_C}{dt} = \boldsymbol{M}_C.$$

Work-energy relation

$$m_i \frac{d\dot{\boldsymbol{r}}_i}{dt} \cdot \frac{\mathrm{d}\boldsymbol{r}_i}{dt} dt = m\dot{\boldsymbol{r}}_i \cdot d\dot{\boldsymbol{r}}_i$$

$$W = \sum_i \int_{(\mathrm{i})}^{(\mathrm{f})} \left(\boldsymbol{F}_i + \sum_{\substack{j \\ j \neq i}} \boldsymbol{F}_{ij} \right) \cdot \mathrm{d}\boldsymbol{r}_i = \sum_i \int_{(\mathrm{i})}^{(\mathrm{f})} m_i \ddot{\boldsymbol{r}}_i \cdot \mathrm{d}\boldsymbol{r}_i$$

resultant force on *i*

$$= \int_{(\mathrm{i})}^{(\mathrm{f})} \mathrm{d}\left(\sum_i \frac{1}{2} m_i \dot{\boldsymbol{r}}_i \cdot \dot{\boldsymbol{r}}_i \right)$$

$$= T(\mathrm{f}) - T(\mathrm{i})$$

$$\boldsymbol{F}_{12}\cdot \mathrm{d}\boldsymbol{r}_1 + \boldsymbol{F}_{21}\cdot \mathrm{d}\boldsymbol{r}_2 = \boldsymbol{F}_{12}\cdot \mathrm{d}\left(\boldsymbol{r}_1 - \boldsymbol{r}_2\right)$$

Conservative force

$$= -\nabla U_{12}\cdot \mathrm{d}\boldsymbol{r}_{12}$$

$$= -\mathrm{d}U_{12}$$

$$U \equiv \frac{1}{2}\sum_{\substack{i,j\\ i\neq j}} U_{ij} = \sum_{\substack{i,j\\ i>j}} U_{ij}$$

$$\sum_i \int_{(\mathrm{i})}^{(\mathrm{f})} \left(\sum_{\substack{j\\ j\neq i}} \boldsymbol{F}_{ij}\right)\cdot \mathrm{d}\boldsymbol{r}_i = U(\mathrm{i}) - U(\mathrm{f})$$

$$W = \sum_i \int_{(\mathrm{i})}^{(\mathrm{f})} \left(\boldsymbol{F}_i + \sum_{\substack{j \\ j \neq i}} \boldsymbol{F}_{ij} \right) \cdot \mathrm{d}\boldsymbol{r}_i$$

$$= W_{\mathrm{ext}} + U(\mathrm{i}) - U(\mathrm{f})$$

$$W_{\mathrm{ext}} = U(\mathrm{f}) - U(\mathrm{i}) + T(\mathrm{f}) - T(\mathrm{i})$$

If part of external force is conservative,

$$W_{\mathrm{ext}} = W_{\mathrm{nonconservative}} + \left(\sum_i U_i \bigg|_{\mathrm{i}} - \sum_i U_i \bigg|_{\mathrm{f}} \right)$$

one-body potential

where U_i is single-body potential corresponding to external force on ith particle.

Now

$$U = \sum_{i} U_i + \frac{1}{2} \sum_{\substack{i,j \\ i \neq j}} U_{ij}$$

$$= \sum_{i} U_i + \sum_{\substack{i,j \\ i > j}} U_{ij}$$

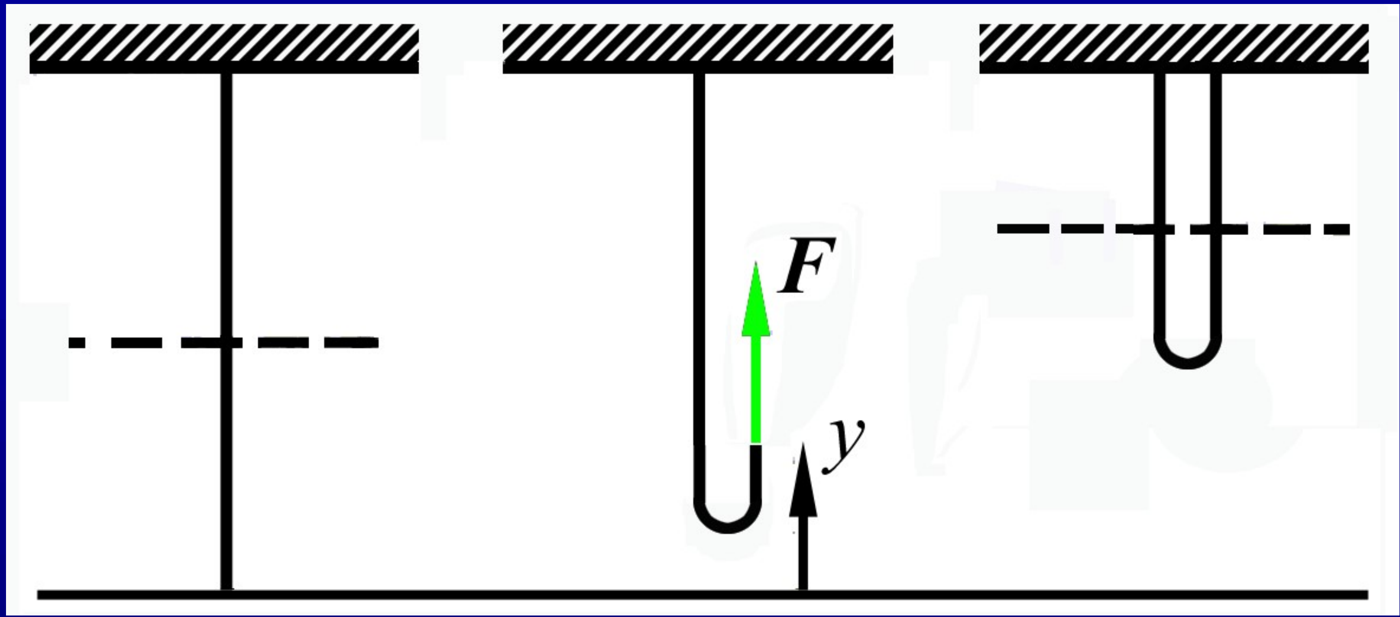
Kinetic energy

$$T = \frac{1}{2}\sum_i m_i\, \dot{\boldsymbol{r}}_i \cdot \dot{\boldsymbol{r}}_i$$

$$= \frac{1}{2}\sum_i m_i \left(\dot{\boldsymbol{r}}_i' + \dot{\boldsymbol{r}}_\mathrm{C}\right)\cdot\left(\dot{\boldsymbol{r}}_i' + \dot{\boldsymbol{r}}_\mathrm{C}\right)$$

$$= \frac{1}{2} m_\mathrm{C} v_\mathrm{C}^2 + \frac{1}{2}\sum_i m_i\, v_i'^2 + \dot{\boldsymbol{r}}_\mathrm{C} \cdot \cancel{\sum_i m_i\, \dot{\boldsymbol{r}}_i'}$$

$$= T_\mathrm{C} + \sum_i T_i'$$



$$F(y) = \frac{1}{2}\left(\frac{m}{\ell}y\right)g$$

$$W = \int_0^{\ell} F(y)\,\mathrm{d}y = \frac{1}{4}mg\ell$$

$$W = mg\left(\frac{3}{4}\ell\right) - mg\left(\frac{1}{2}\ell\right) = \frac{1}{4}mg\ell$$

Scissor kick 剪

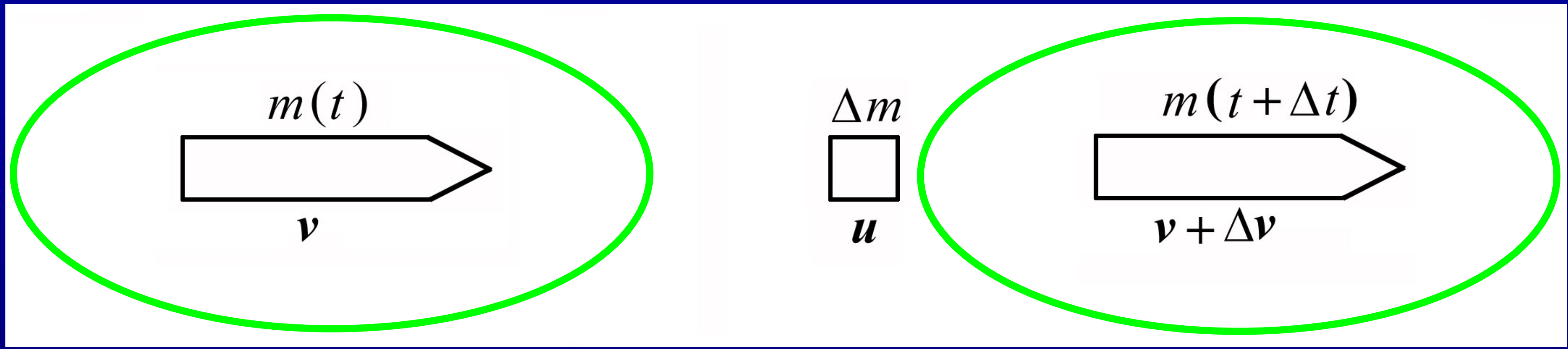
Straddle jump 俯卧式

Fosbury flop 背跃

Dick Fosbury using the "Fosbury Flop," a then-unorthodox head-first, back-to-the-bar method of high jumping, at the Mexico City Games. He cleared 7 feet 41/4 inches (2.24 meters) for a gold medal and a world record on October 20, 1968 .

Gold medal winner Ethel Catherwood of Canada *scissors* over the bar at the 1928 Summer Olympics. Her winning result was 1.59 m.

*5.2 System of variable mass



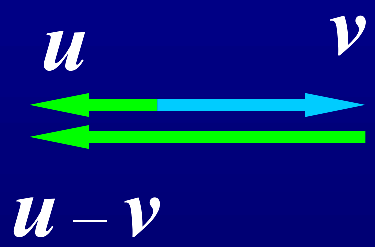
$$\frac{\Delta \boldsymbol{p}}{\Delta t}=\frac{\left[(m-\Delta m)(\boldsymbol{v}+\Delta \boldsymbol{v})+(\Delta m)\boldsymbol{u}\right]-m\boldsymbol{v}}{\Delta t}$$

$$=m\frac{\Delta \boldsymbol{v}}{\Delta t}+\left[\boldsymbol{u}-(\boldsymbol{v}+\cancel{\Delta \boldsymbol{v}})\right]\frac{\Delta m}{\Delta t}$$

$$\frac{\mathrm{d}\boldsymbol{p}}{\mathrm{d}t}=m\frac{\mathrm{d}\boldsymbol{v}}{\mathrm{d}t}+(\boldsymbol{u}-\boldsymbol{v})\left(-\frac{\mathrm{d}m}{\mathrm{d}t}\right)$$

$$\mathrm{d}m=-\Delta m$$

$$m\frac{\mathrm{d}\boldsymbol{v}}{\mathrm{d}t}=\boldsymbol{F}_{\mathrm{ext}}+\left(\boldsymbol{u}-\boldsymbol{v}\right)\frac{\mathrm{d}m}{\mathrm{d}t}$$



$|\boldsymbol{u}-\boldsymbol{v}| = v_{\mathrm{rel}}$

exhaust speed 喷气速度

Example 5.4 $m_0 = 2.72\times10^6\,\text{kg} \quad m_\text{f} = 2.52\times10^6\,\text{kg}$

$$-\frac{\mathrm{d}m}{\mathrm{d}t} = C = 1.29\times10^3\,\text{kg/s}$$

$$v_\text{rel} = 5.50\times10^4\,\text{m/s}$$

$$m\frac{\mathrm{d}\boldsymbol{v}}{\mathrm{d}t} = m\boldsymbol{g} + \boldsymbol{v}_\text{rel}\frac{\mathrm{d}m}{\mathrm{d}t}$$

$$\boldsymbol{g} = -g\boldsymbol{k}$$

$$\boldsymbol{v} = v\boldsymbol{k}$$

$$\frac{\mathrm{d}v}{\mathrm{d}t} + \frac{v_\text{rel}}{m}\frac{\mathrm{d}m}{\mathrm{d}t} = -g$$

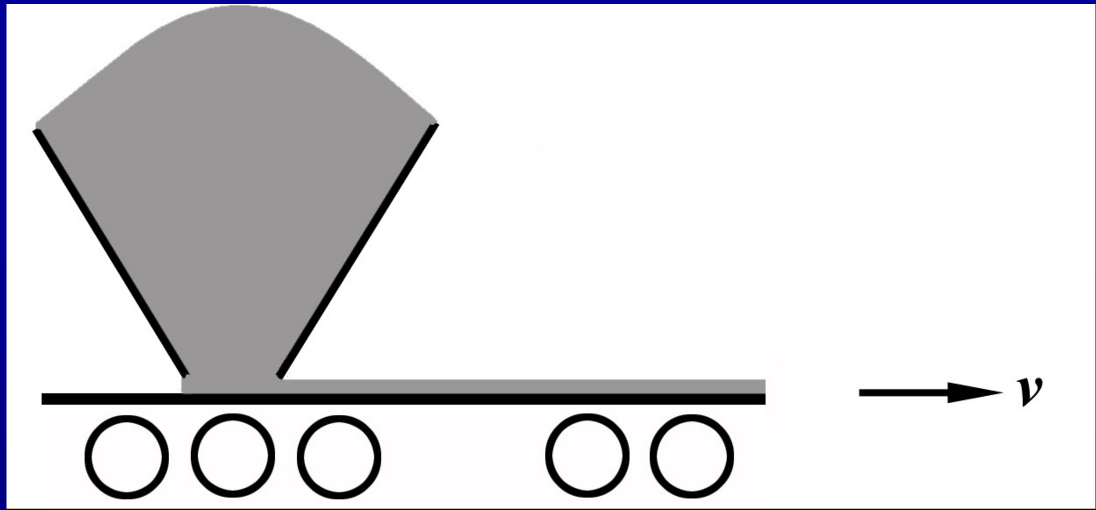
$$\boldsymbol{v}_\text{rel} = -v_\text{rel}\boldsymbol{k}$$

$$\int_{v_0}^{v_f} \mathrm{d}v + v_{\mathrm{rel}} \int_{m_0}^{m_f} \frac{\mathrm{d}m}{m} = -g \int_{0}^{t} \mathrm{d}t$$

$$v_f = v_0 + v_{\mathrm{rel}} \ln\left(\frac{m_0}{m_f}\right) - gt$$

$$= 2.68 \times 10^3 \ \mathrm{m/s}$$

Example 5.5



$$u = 0; \qquad v = \text{const.}$$

$$0 = F_{\text{ext}} + (0 - v)\frac{\mathrm{d}m}{\mathrm{d}t}$$

The power is

$$P = F_{\text{ext}} \cdot v = v\frac{\mathrm{d}m}{\mathrm{d}t} \cdot v \overset{?}{=} 2\frac{\mathrm{d}}{\mathrm{d}t}\left(\frac{1}{2}mv^2\right)$$

In dt **, when dropped on the conveyer belt,** dm **is accelerated along it from zero to velocity** v**. The average acceleration is** $(v-0)/\mathrm{d}t$**. During accelerating, sand slides a distance:**

$$\mathrm{d}s=\frac{1}{2}a(\mathrm{d}t)^2=\frac{1}{2}\frac{v}{\mathrm{d}t}(\mathrm{d}t)^2=\frac{1}{2}v\mathrm{d}t$$

The frictional force does work

$$\mathrm{d}W=F'\cdot\mathrm{d}s=\frac{1}{2}F'\cdot v\mathrm{d}t=\frac{1}{2}F_{\mathrm{ext}}\cdot v\mathrm{d}t$$

Assignment: 5.1, 5.2, *5.3

http://www.physics.harvard.edu/academics/undergrad/problems.html

5.3 Collisions

Possible definitions of collision:

- **a close encounter**
- **a process of interaction with impulsive force**
- **a process during which two particles (systems) exchange their momentum and energy**

  with short range force and

  clear-cut duration

e.g. Lennard−Jones potential (6-12 potential)

$$U = 4\varepsilon\left[\left(\frac{\sigma}{r}\right)^{12} - \left(\frac{\sigma}{r}\right)^{6}\right]$$

Coulomb scattering is a special topic.

Gravitational ~

Collisions in microscopic world:

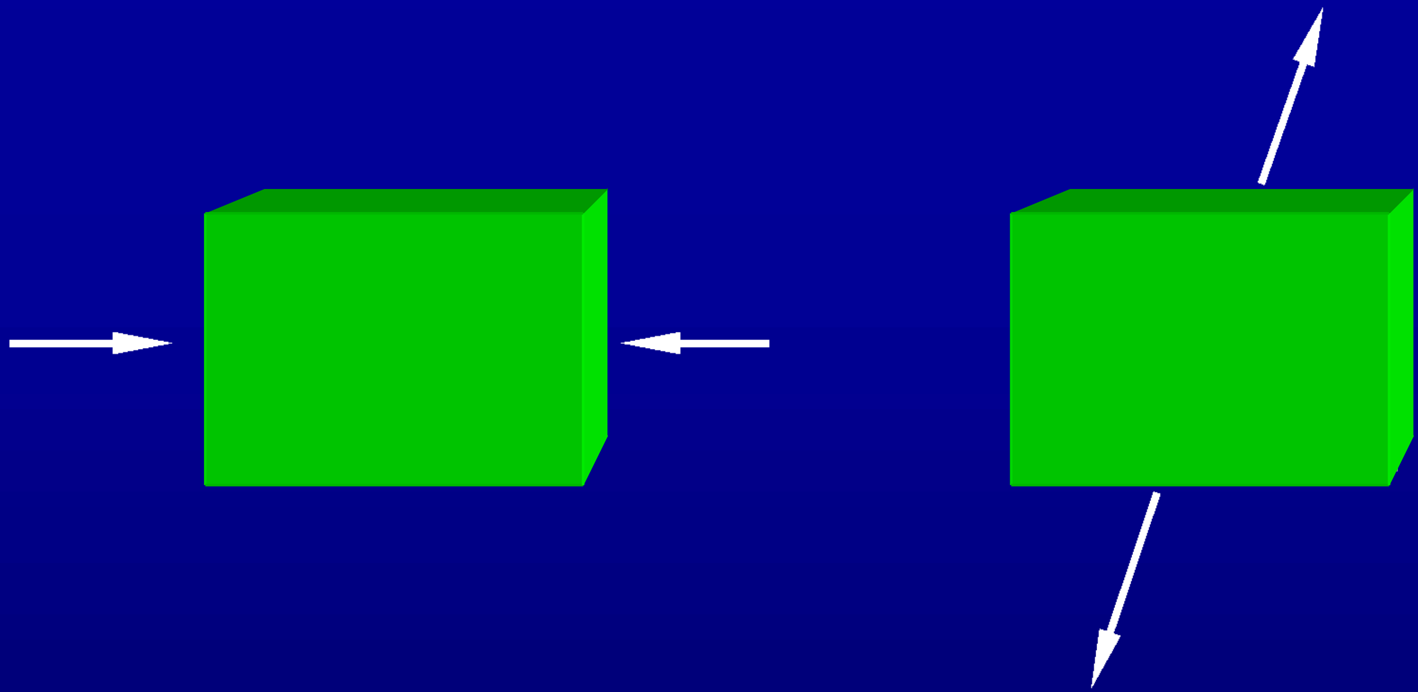
$$\alpha + \mathrm{Au} \longrightarrow \alpha + \mathrm{Au}$$ **(Rutherford scattering)**

$$\mathrm{n}_{\mathrm{fast}} + {}^{2}\mathrm{H} \longrightarrow \mathrm{n}_{\mathrm{slow}} + {}^{2}\mathrm{H}$$ **(modulator)**

$${}^{2}\mathrm{H} + {}^{2}\mathrm{H} \longrightarrow {}^{3}\mathrm{He} + \mathrm{n} + 3.27\mathrm{MeV}$$ **(fusion)**

$$\mathrm{n} \longrightarrow \mathrm{p} + \mathrm{e}^{-} + \bar{\nu}_{\mathrm{e}}$$ **(β decay)**

$$\mathrm{e}^{-} + \mathrm{p} \longrightarrow \mathrm{n} + \nu_{\mathrm{e}} \quad (E_{\nu} = cp_{\nu})$$



initial state *final state*

$i \rightleftarrows f$ ("black box")

known $i, f \longrightarrow$ black box

(inverse scattering method)

reverse engineering proprietary

$$\boldsymbol{p}_{\mathrm{i}} = \boldsymbol{p}_{\mathrm{f}}$$

$$T_{\mathrm{i}} + U_{\mathrm{i}} = T_{\mathrm{f}} + U_{\mathrm{f}}$$

Define Q **"reaction heat"**

$$Q \equiv T_{\mathrm{f}} - T_{\mathrm{i}} = U_{\mathrm{i}} - U_{\mathrm{f}}$$

potential energy or internal energy

$Q = 0$ **elastic collision**

$Q < 0$ **endoergic (endothermic) reaction** 吸能(热)反应

$Q > 0$ **exoergic (exothermic) reaction, fusion, fission, explosion**

$Q = 0$ head-on collision

$m_1 v_1 \quad m_2 v_2 \qquad\qquad m_1 v_1' \quad m_2 v_2'$

before collision **after collision**

$$v_{\text{app}} \equiv v_1 - v_2 \,, \qquad v_{\text{sep}} \equiv v_2' - v_1'$$

$$\begin{cases} m_1 v_1 + m_2 v_2 = m_1 v_1' + m_2 v_2' \\ Q + \dfrac{1}{2} m_1 \, v_1^2 + \dfrac{1}{2} m_2 \, v_2^2 = \dfrac{1}{2} m_1 \, {v_1'}^2 + \dfrac{1}{2} m_2 \, {v_2'}^2 \end{cases}$$

Pay some attention to its position

$Q = 0$ head-on collision

$m_1 v_1 \quad m_2 v_2 \qquad\qquad m_1 v_1' \quad m_2 v_2'$

before collision **after collision**

$$\boldsymbol{v}_{\text{app}} \equiv \boldsymbol{v}_1 - \boldsymbol{v}_2 \,, \qquad \boldsymbol{v}_{\text{sep}} \equiv \boldsymbol{v}_2' - \boldsymbol{v}_1'$$

$$\begin{cases} m_1 v_1 + m_2 v_2 = m_1 v_1' + m_2 v_2' \\ Q + \dfrac{1}{2} m_1 v_1^2 + \dfrac{1}{2} m_2 v_2^2 = \dfrac{1}{2} m_1 v_1'^2 + \dfrac{1}{2} m_2 v_2'^2 \end{cases}$$

$${}^2\text{H} + {}^2\text{H} \longrightarrow {}^3\text{He} + \text{n} \ +3.27\text{MeV}$$

$$\begin{cases} m_1\left(v_1 - v_1'\right) = m_2\left(v_2' - v_2\right) \\ m_1\left(v_1^2 - v_1'^2\right) = m_2\left(v_2'^2 - v_2^2\right) \end{cases} \qquad v_1 \neq v_1'$$

$$v_1 + v_1' = v_2' + v_2 \qquad \textbf{i.e.} \quad v_{\text{app}} = v_{\text{sep}}$$

Set $v_2 = 0$ (for the sake of simplicity)

$$v_1' = \frac{\left(m_1 - m_2\right)v_1 + \cancel{2m_2 v_2}}{m_1 + m_2}$$

$$v_2' = \frac{\cancel{\left(m_2 - m_1\right)v_2} + 2m_1 v_1}{m_1 + m_2}$$

$$v_1' = \frac{m_1 - m_2}{m_1 + m_2} v_1 , \qquad v_2' = \frac{2m_1}{m_1 + m_2} v_1$$

(1) recoil $m_1 << m_2$

$$\begin{cases} v_1' = -\left(1 - 2\dfrac{m_1}{m_2}\right) v_1 \approx -v_1 \\ v_2' = 2\dfrac{m_1}{m_2} v_1 << v_1 \end{cases}$$

$$\frac{T_2'}{T_1'} \approx \frac{T_2'}{T_1} = 4\frac{m_1}{m_2} << 1$$

(2) transfer $m_1 = m_2$

$$v_1' = 0 \ , \quad v_2' = v_1$$

neutron moderator

(3) if $m_1 >> m_2$

$$v_1' \approx \left(1 - 2\frac{m_2}{m_1}\right) v_1 \approx v_1$$

$$v_2' \approx 2\left(1 - \frac{m_2}{m_1}\right) v_1 \approx 2v_1$$

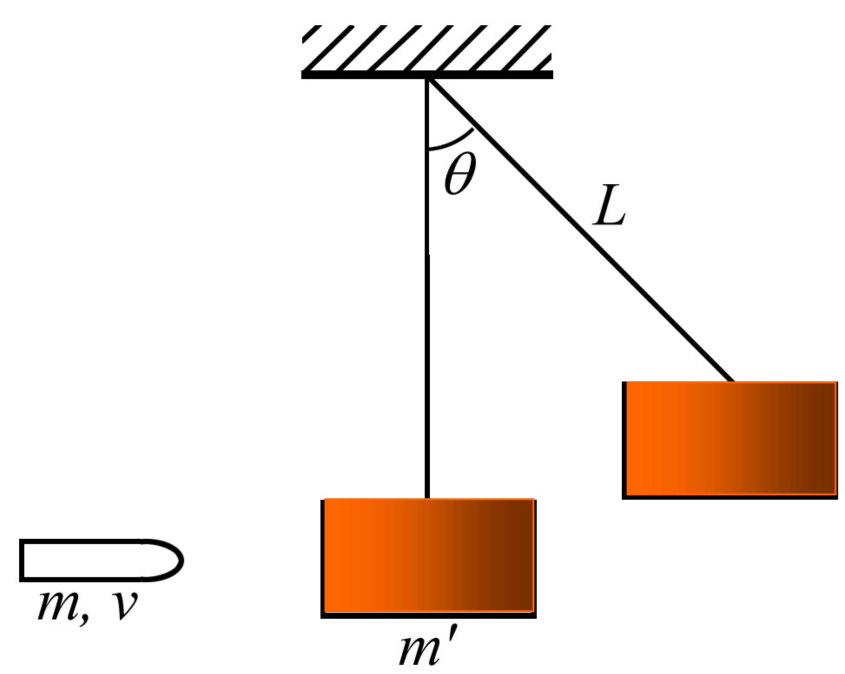
$Q < 0 \ (T_\mathrm{f} < T_\mathrm{i})$ **inelastic collisions** $v_\mathrm{app} > v_\mathrm{sep}$

Define the coefficient of restitution *e* through

$$e(v_1 - v_2) = v_2' - v_1' \quad (0 \le e < 1)$$

$e = 0$ $\longrightarrow$ **Perfectly (or completely) inelastic**

Example 5.6 ballistic pendulum



The conservation of momentum reads

$$mv = (m' + m)v'$$

(before and after hitting)

The conservation of mechanical energy is

$$\frac{1}{2}(m' + m)v'^2 = (m' + m)gL(1 - \cos\theta)$$

(for $\theta = 0$ and θ)

the velocity of bullet v:

$$v = \frac{m' + m}{m}\left[2gL(1 - \cos\theta)\right]^{1/2}$$

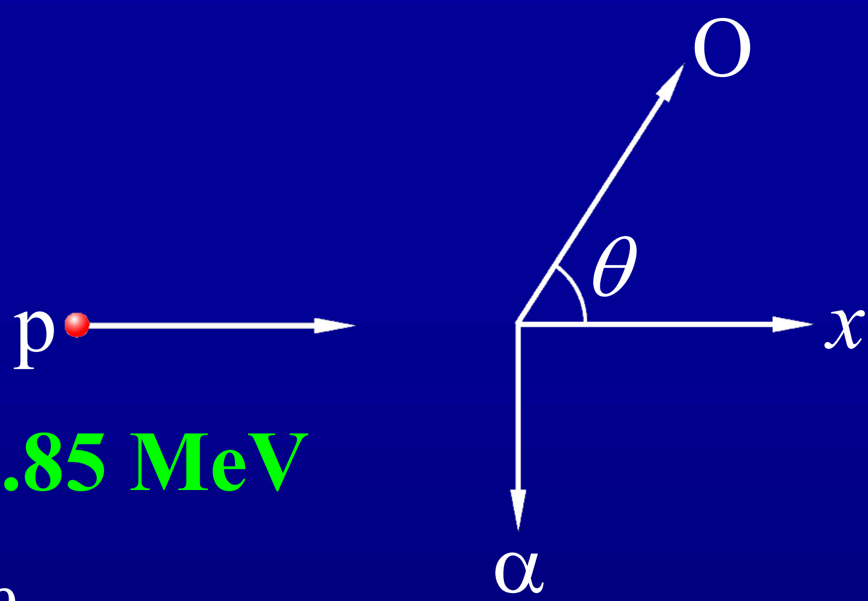
The "reaction heat" is

$$Q = T_{\mathrm{f}} - T_{\mathrm{i}} = \frac{1}{2}(m' + m)v'^2 - \frac{1}{2}mv^2$$

$$= -\frac{m' + m}{m}m'gL(1 - \cos\theta)$$

Example 5.7

$$ {}^{19}\mathrm{F} + \mathrm{p} \longrightarrow {}^{16}\mathrm{O} + \alpha + Q $$



1.95×10^{7} m/s

$$ x: \quad m_{\mathrm{p}}\, v_{\mathrm{p}} = m_{\mathrm{O}}\, v_{\mathrm{O}} \cos\theta $$

$$ y: \quad 0 = m_{\mathrm{O}} v_{\mathrm{O}} \sin\theta - m_{\alpha} v_{\alpha} $$

fluorine 氟

$$ 1.85\mathrm{MeV} << 938\mathrm{MeV} = m_{\mathrm{p}} c^{2} $$

$$ \left(m_{\mathrm{O}} v_{\mathrm{O}}\right)^{2} = \left(m_{\mathrm{p}} v_{\mathrm{p}}\right)^{2} + \left(m_{\alpha} v_{\alpha}\right)^{2} $$

$$ Q + \frac{1}{2} m_{\mathrm{p}}\, v_{\mathrm{p}}^{2} = \frac{1}{2} m_{\mathrm{O}}\, v_{\mathrm{O}}^{2} + \frac{1}{2} m_{\alpha}\, v_{\alpha}^{2} $$

Reaction energy is

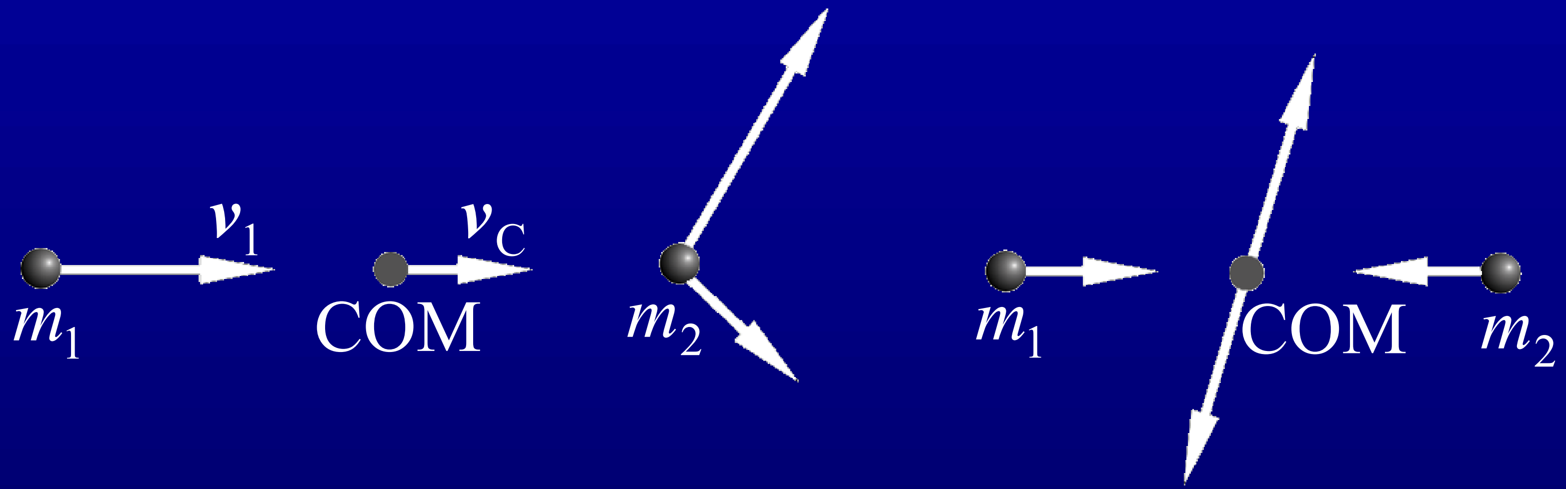
$$Q=\frac{1}{2}\left[\frac{1}{m_{\mathrm{O}}}\left(m_{\mathrm{O}}\,v_{\mathrm{O}}\right)^2+\frac{1}{m_{\alpha}}\left(m_{\alpha}\,v_{\alpha}\right)^2-\frac{1}{m_{\mathrm{p}}}\left(m_{\mathrm{p}}\,v_{\mathrm{p}}\right)^2\right]$$

$$=\frac{1}{2}\left[\frac{1}{m_{\mathrm{O}}}\left(m_{\mathrm{p}}\,v_{\mathrm{p}}\right)^2+\frac{1}{m_{\mathrm{O}}}\left(m_{\alpha}\,v_{\alpha}\right)^2\right.$$

$$\left.+\frac{1}{m_{\alpha}}\left(m_{\alpha}\,v_{\alpha}\right)^2-\frac{1}{m_{\mathrm{p}}}\left(m_{\mathrm{p}}\,v_{\mathrm{p}}\right)^2\right]$$

$$=\left(1-\frac{m_{\alpha}}{m_{\mathrm{O}}}\right)T_{\alpha}-\left(1-\frac{m_{\mathrm{p}}}{m_{\mathrm{O}}}\right)T_{\mathrm{p}} \qquad Q=8.10\mathrm{MeV}$$

L-frame and C-frame



$$\boldsymbol{p} = m_1\, \boldsymbol{v}_1$$

$$\boldsymbol{v}_C = \frac{m_1}{m_1 + m_2}\boldsymbol{v}_1 \quad \text{or} \quad \boldsymbol{p}_C = (m_1 + m_2)\boldsymbol{v}_C = m_1\, \boldsymbol{v}_1 = \boldsymbol{p}$$

In C-frame

$$\boldsymbol{v}_{1\mathrm{C}} = \boldsymbol{v}_1 - \boldsymbol{v}_\mathrm{C} = \frac{m_2}{m_1 + m_2}\boldsymbol{v}_1, \quad \boldsymbol{v}_{2\mathrm{C}} = \mathbf{0} - \boldsymbol{v}_\mathrm{C} = -\frac{m_1}{m_1 + m_2}\boldsymbol{v}_1$$

$$\boldsymbol{p}_{1\mathrm{C}} = m_1\,\boldsymbol{v}_{1\mathrm{C}} = \mu\boldsymbol{v}_1, \qquad \boldsymbol{p}_{2\mathrm{C}} = m_2\,\boldsymbol{v}_{2\mathrm{C}} = -\mu\boldsymbol{v}_1$$

The conservation momentum reads

$$m_1\,\boldsymbol{v}_{1\mathrm{C}} + m_2\,\boldsymbol{v}_{2\mathrm{C}} = 0, \quad m_1\,\boldsymbol{v}'_{1\mathrm{C}} + m_2\,\boldsymbol{v}'_{2\mathrm{C}} = 0$$

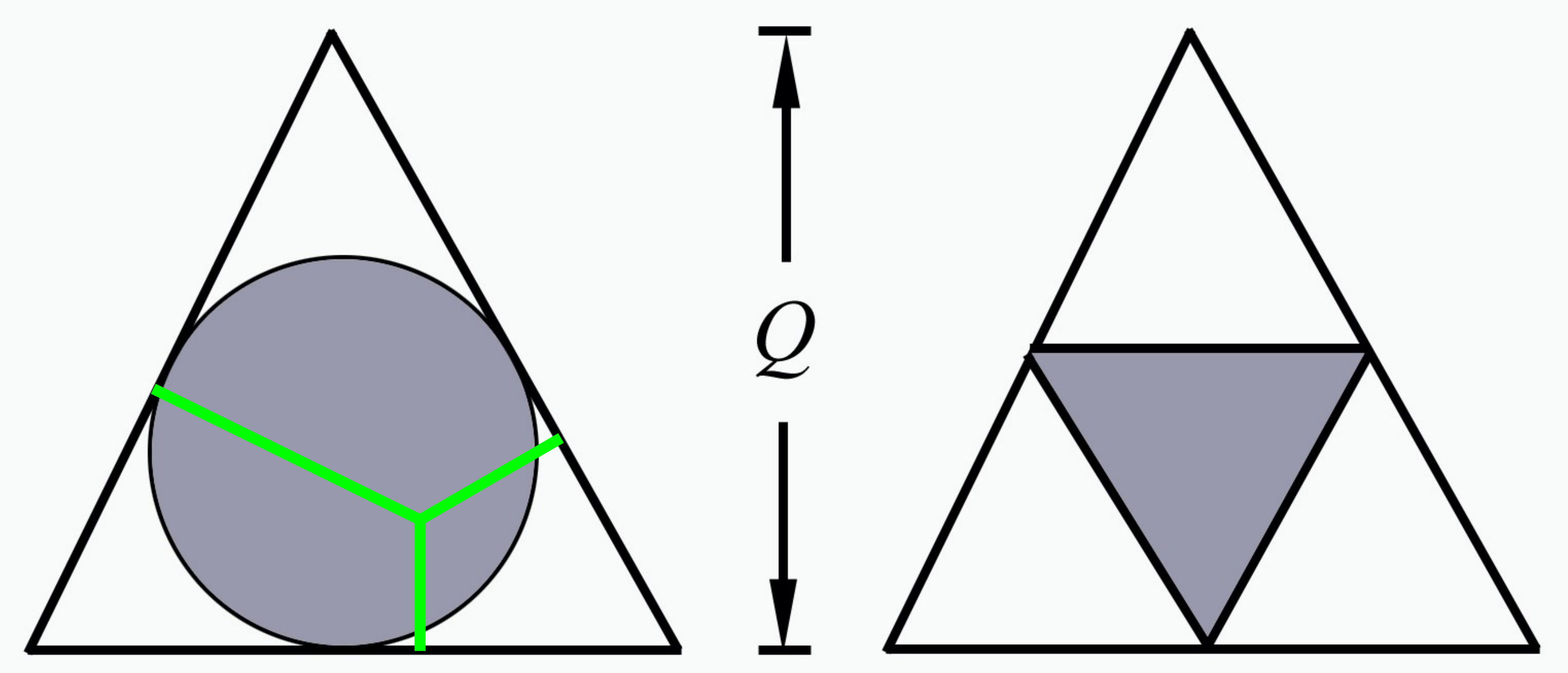
The conservation of energy reads

$$Q + \frac{1}{2}m_1\,v_{1\mathrm{C}}^2 + \frac{1}{2}m_2\,v_{2\mathrm{C}}^2 = \frac{1}{2}m_1\,v_{1\mathrm{C}}'^2 + \frac{1}{2}m_2\,v_{2\mathrm{C}}'^2$$

$$Q = 0 \longrightarrow v_{1\mathrm{C}} = v'_{1\mathrm{C}}\,, \quad v_{2\mathrm{C}} = v'_{2\mathrm{C}}$$

*Dalitz plot

$$m_0 \longrightarrow 3m + Q$$

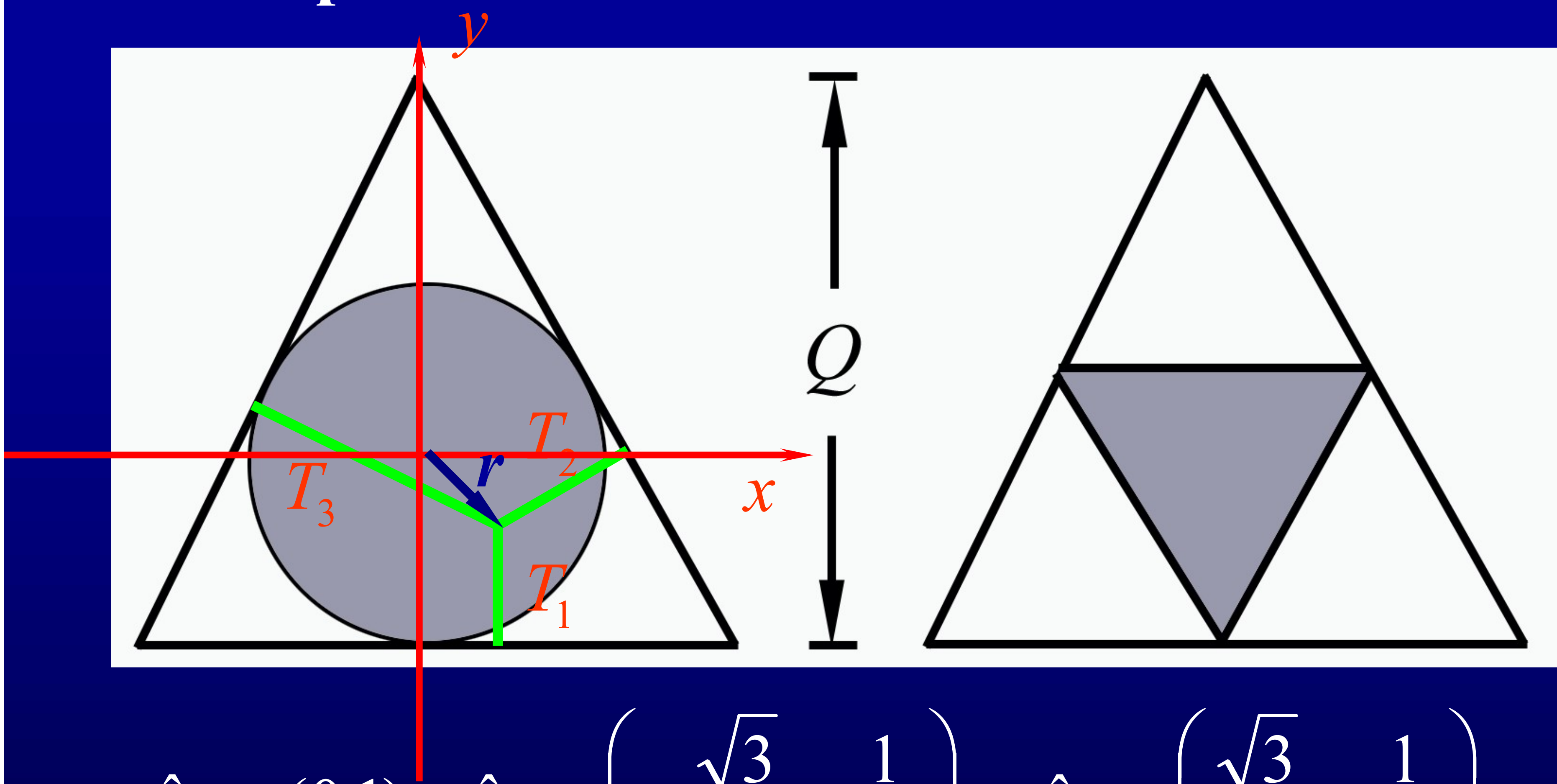


$$E = \frac{p^2}{2m}$$

$$E = cp$$

Energy-momentum relation

Dalitz plot



$$\hat{\mathbf{n}}_1 = (0,1), \quad \hat{\mathbf{n}}_2 = \left(-\frac{\sqrt{3}}{2}, -\frac{1}{2}\right), \quad \hat{\mathbf{n}}_3 = \left(\frac{\sqrt{3}}{2}, -\frac{1}{2}\right)$$

$$T_i = \frac{Q}{3} + \boldsymbol{r} \cdot \hat{\boldsymbol{n}}_i \qquad i = 1,\ 2,\ 3$$

Conservation of momentum

$$\sum_{i=1,2,3} \boldsymbol{p}_i = 0 \qquad \boldsymbol{p}_1 = -\boldsymbol{p}_2 - \boldsymbol{p}_3$$

$$p_1^2 = p_2^2 + p_3^2 + 2 p_2 p_3 \cos\theta_{23}$$

$$T_i = \frac{p_i^2}{2m}$$

$$T_1 = T_2 + T_3 + 2\sqrt{T_2 T_3}\cos\theta_{23}$$

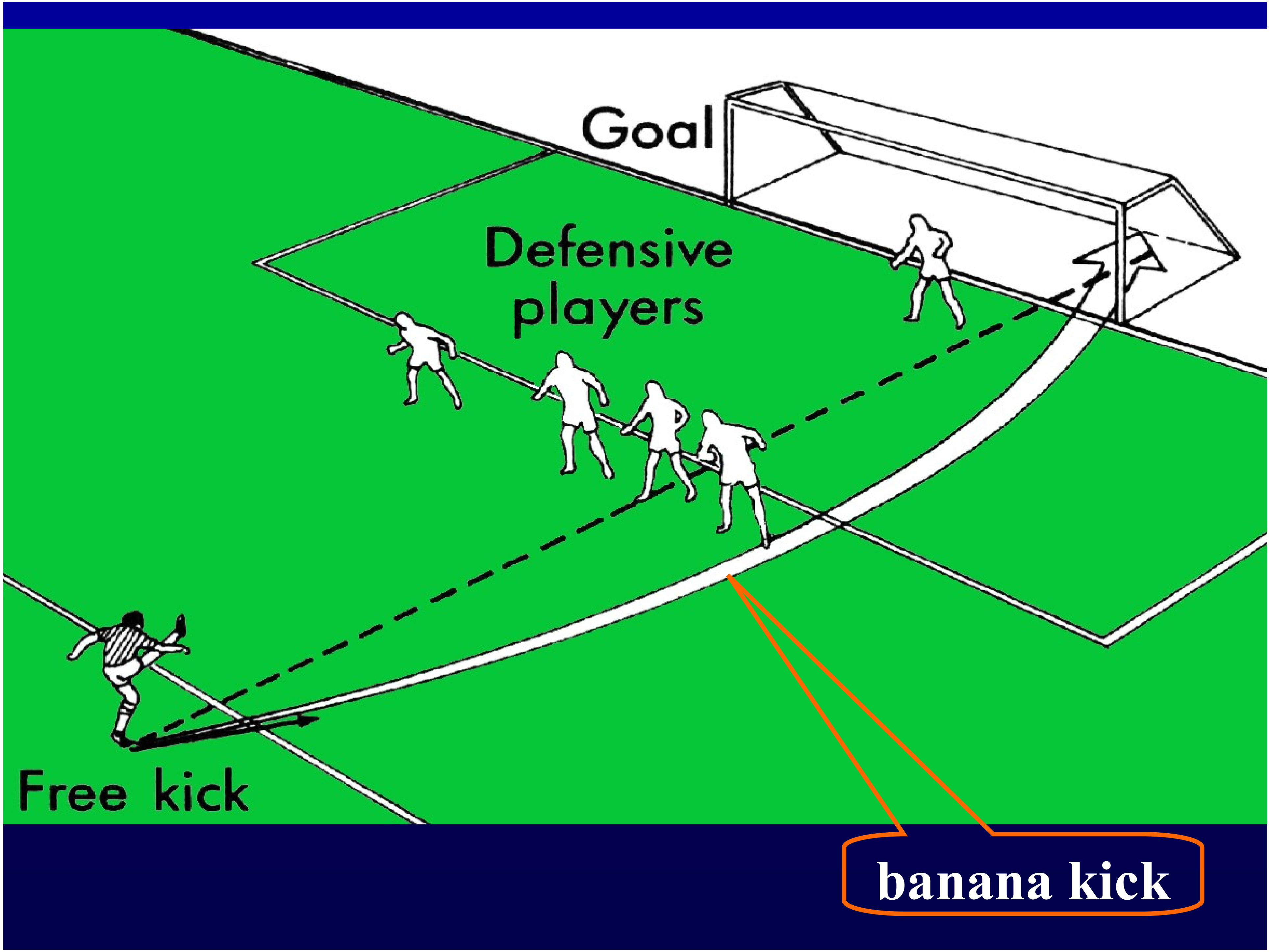
One can see that T_1 is maximum when $\cos\theta_{23} = 1.$

$$\frac{Q}{3}+y=\left(\frac{Q}{3}-\frac{\sqrt{3}}{2}x-\frac{1}{2}y\right)+\left(\frac{Q}{3}+\frac{\sqrt{3}}{2}x-\frac{1}{2}y\right)$$

$$+2\sqrt{\left(\frac{Q}{3}-\frac{\sqrt{3}}{2}x-\frac{1}{2}y\right)\left(\frac{Q}{3}+\frac{\sqrt{3}}{2}x-\frac{1}{2}y\right)}$$

$$x^2+y^2=\left(\frac{Q}{3}\right)^2$$

Assignment :5.6, 5.11


Goal
Defensive
players
Free kick
banana kick

Roberto Carlos' banana free kick

Brazil vs France

Le Tournoi, Lyons, 3 June, 1997, 22nd Minute of the Match

This is probably one of the best free kicks ever recorded.

5.4 Fluid motion

Fluid is a continuum.

We may image it is composed of fluid elements —fluid particles. Their motion could be described as for other many-particle system

This method is deserved to J L Lagrange (luh 'grah(n)zh).

According to L Euler ('oy ler rhymes with toiler) we can study instead the distributions of density and velocity in the space the fluid situates and their variation with time:

$$\rho = \rho(r,t), \quad v = v(r,t)$$

Fluid could be

steady	or	nonsteady	(tide, turbulent)
irrotational		rotational	(whirlpool)
without		with source and/or sink	
incompressible		compressible	
nonviscous		viscous	

We take the simplest model.

ideal fluid : incompressible fluid without viscosity

Laminar and Turbulent Flows

The plume from this candle flame goes from laminar to turbulent.

Features of a turbulence:

Irregularity

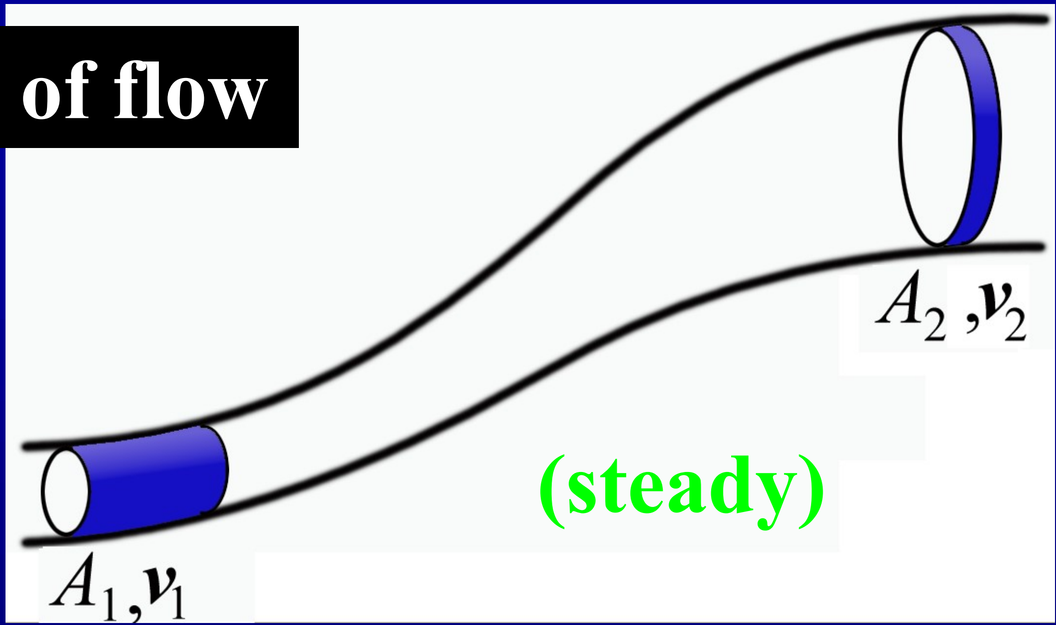
Diffusivity

Rotationality

Dissipation

Richard Feynman : Turbulence is "the most important unsolved problem of classical physics."

Streamline & tube of flow



$$\Delta m_1 = \rho_1 A_1 v_1 \Delta t, \quad \Delta m_2 = \rho_2 A_2 v_2 \Delta t$$

$$\frac{\mathrm{d}m}{\mathrm{d}t} = \text{constant}$$ **(without source and sink)**

$$\rho_1 A_1 v_1 = \rho_2 A_2 v_2 \quad \rho A v = \text{constant}$$

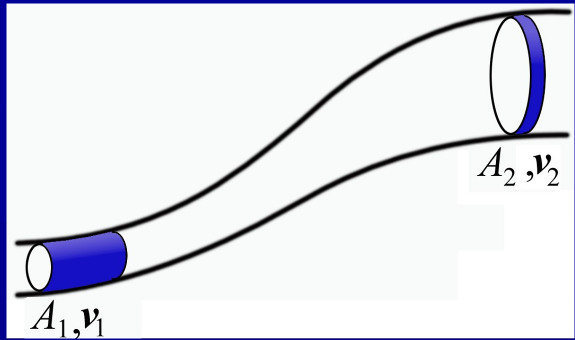
(incompressible) $Av = \text{constant}$

equation of continuity

Work-energy relation

$$W = W_{\mathrm{p}} + W_{\mathrm{g}} = \Delta T\,, \quad W_{\mathrm{p}} - \Delta U = \Delta T$$

$$W_{\mathrm{p}} = \Delta T + \Delta U$$



While

$$W_{\mathrm{p}} = p_1 A_1 (v_1 \Delta t) - p_2 A_2 (v_2 \Delta t) = (p_1 - p_2) \Delta V$$

$$\Delta T = \frac{1}{2} \Delta m \left(v_2^2 - v_1^2\right) \qquad \Delta U = \Delta m g (z_2 - z_1)$$

we have

$$p_1 - p_2 = \frac{1}{2} \frac{\Delta m}{\Delta V} \left(v_2^2 - v_1^2\right) + \frac{\Delta m}{\Delta V} g (z_2 - z_1)$$

Notice

$$\lim_{\Delta V \to 0} \frac{\Delta m}{\Delta V} = \rho$$

$$p_1 + \frac{1}{2}\rho v_1^2 + \rho g z_1 = p_2 + \frac{1}{2}\rho v_2^2 + \rho g z_2$$

we have Bernoulli's equation

$$p + \frac{1}{2}\rho\, v^2 + \rho\, g\, z = \text{constant}$$

valid along streamline or tube of flow

- **static equilibrium** $p + \rho\, g\, z = \text{constant}$

*Euler equation $\dfrac{\mathrm{d}\boldsymbol{v}}{\mathrm{d}t} = -\dfrac{1}{\rho}\nabla p + \boldsymbol{g}$

static $\nabla p = \rho \boldsymbol{g}$, $\dfrac{\mathrm{d}p}{\mathrm{d}z}\boldsymbol{k} = -\rho g \boldsymbol{k}$

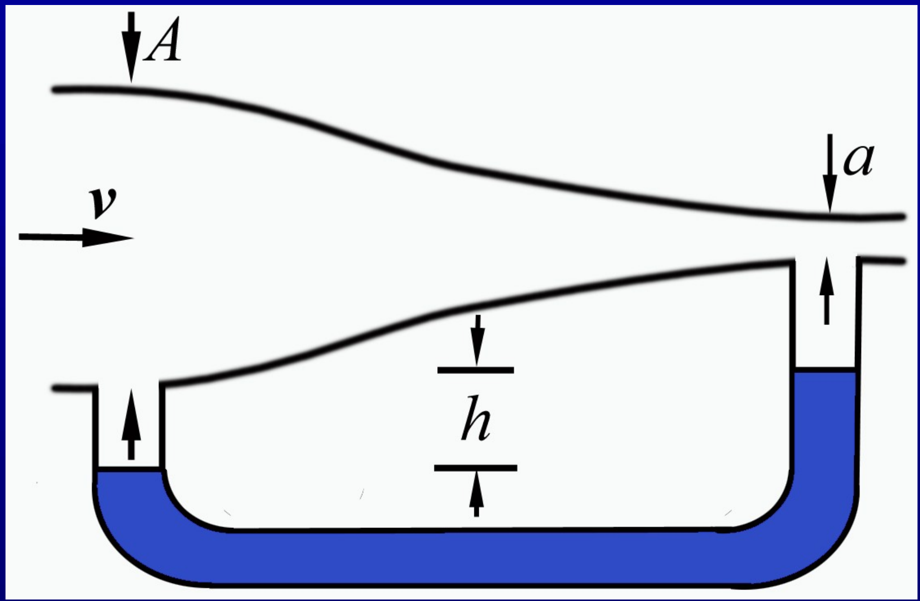
- z **= constant** $p + \dfrac{1}{2}\rho\, v^2 = \text{constant}$

$$\boldsymbol{v} = \boldsymbol{v}(\boldsymbol{r}, t)$$

$$\frac{dv_x}{dt} = \frac{\partial v_x}{\partial t} + \frac{\partial v_x}{\partial x}\frac{dx}{dt} + \frac{\partial v_x}{\partial y}\frac{dy}{dt} + \frac{\partial v_x}{\partial z}\frac{dz}{dt}$$

$$= \frac{\partial v_x}{\partial t} + v_x\frac{\partial v_x}{\partial x} + v_y\frac{\partial v_x}{\partial y} + v_z\frac{\partial v_x}{\partial z}$$

Application: Venturi meter



$$Av = av_a$$

(continuity)

$$p_A + \frac{1}{2}\rho v^2 = p_a + \frac{1}{2}\rho v_a^2$$

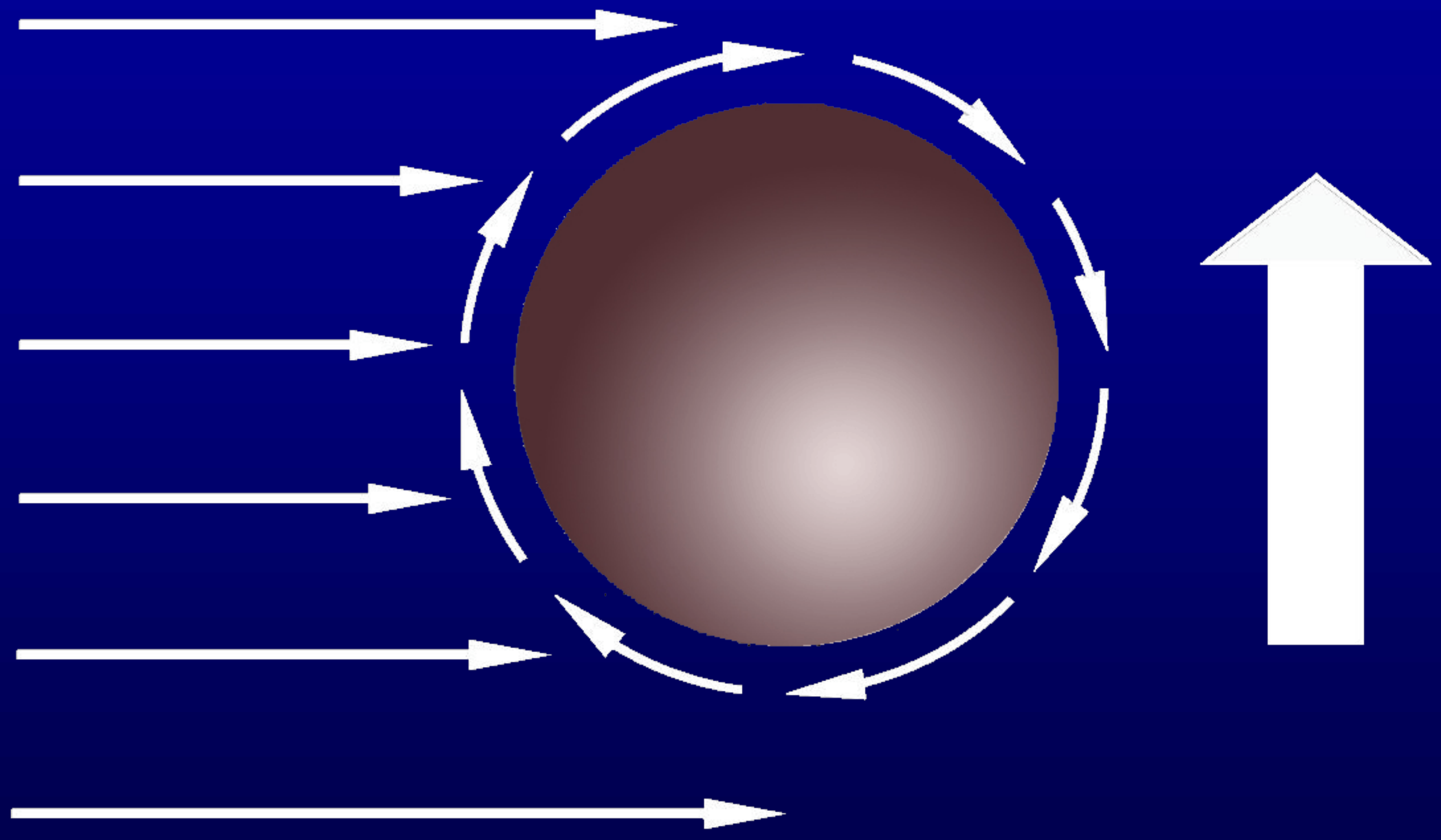
(horizontal)

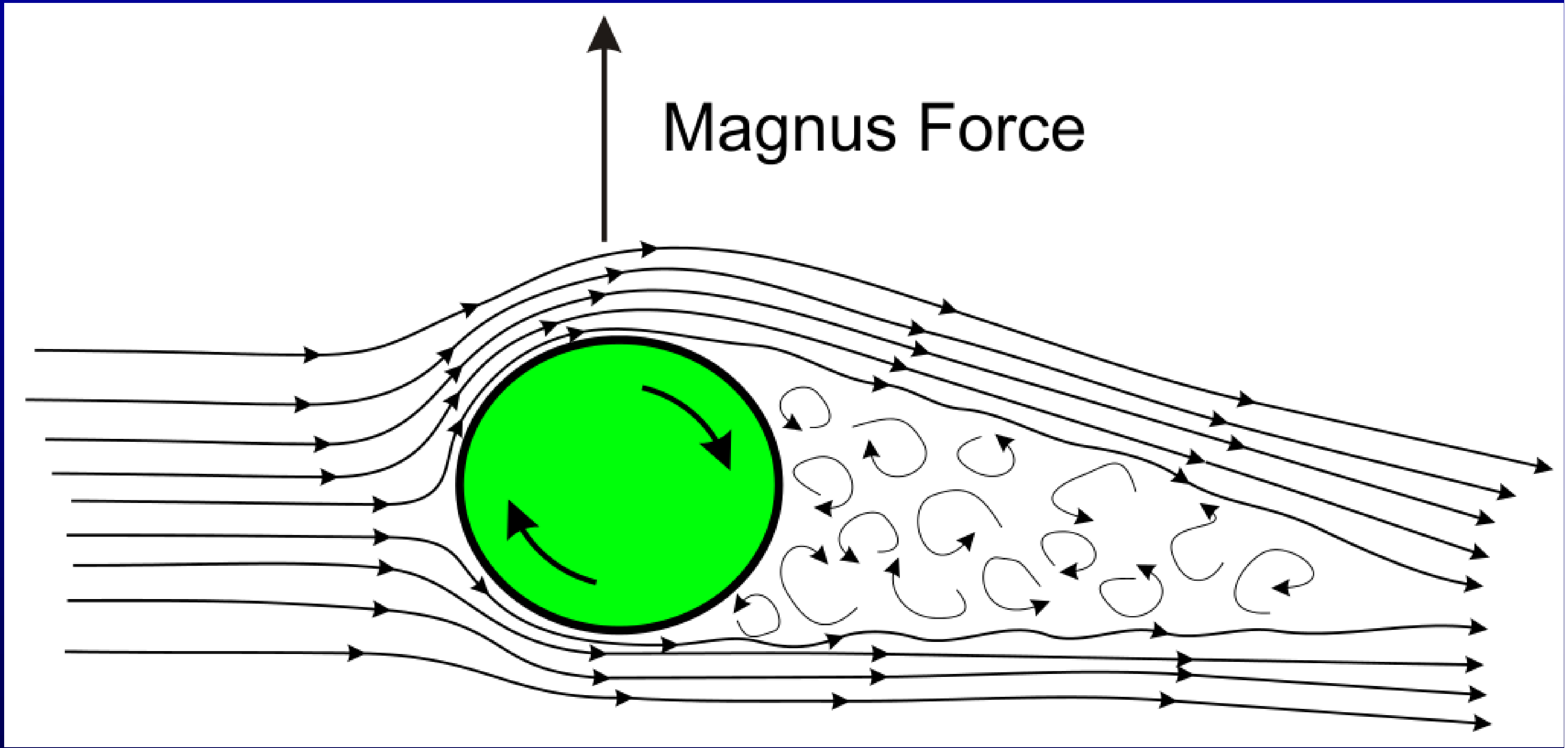
$$p_A (+\rho\, g\, h) = p_a + \rho_{\mathrm{Hg}}\, g\, h$$

static(for U-tube)

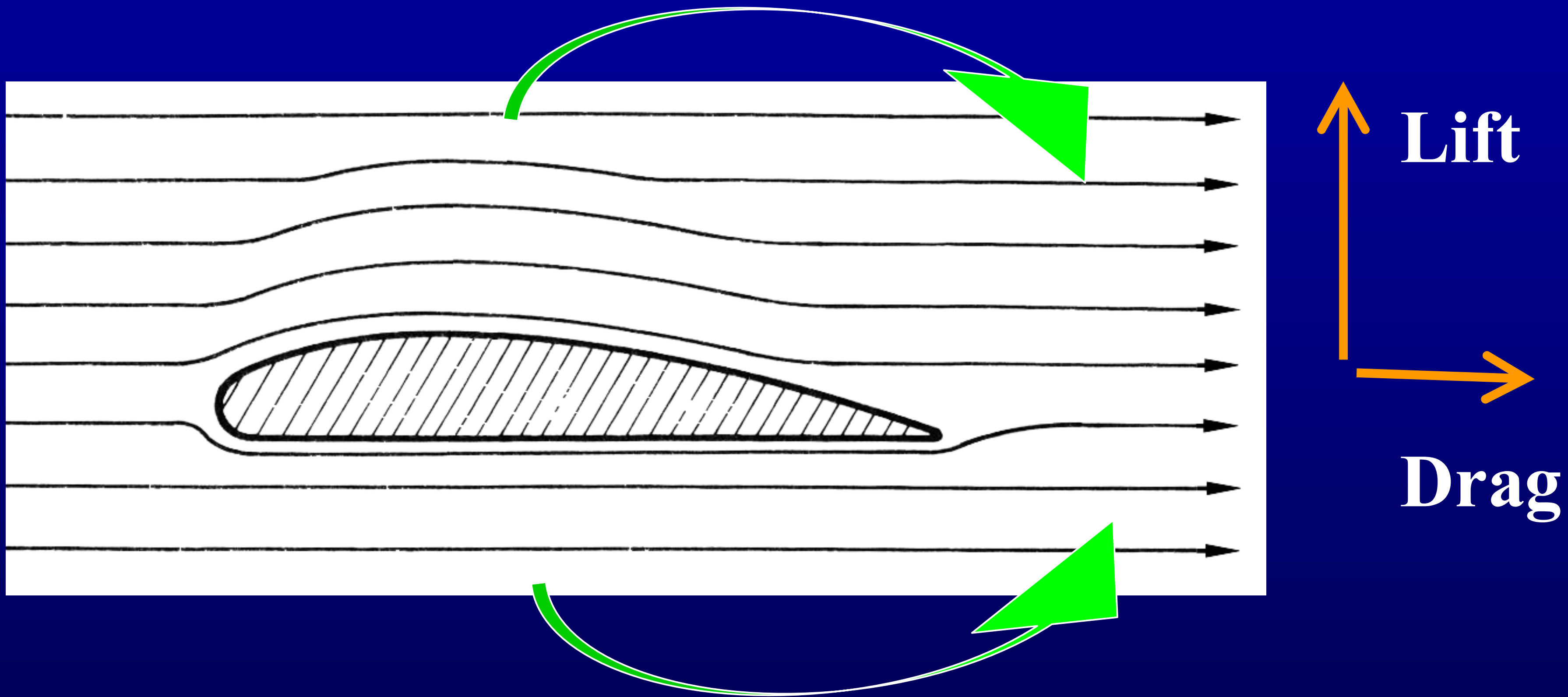
So Venturi meter can be used to measure *v* by calibrating *h*.

Application: **Magnus effect**

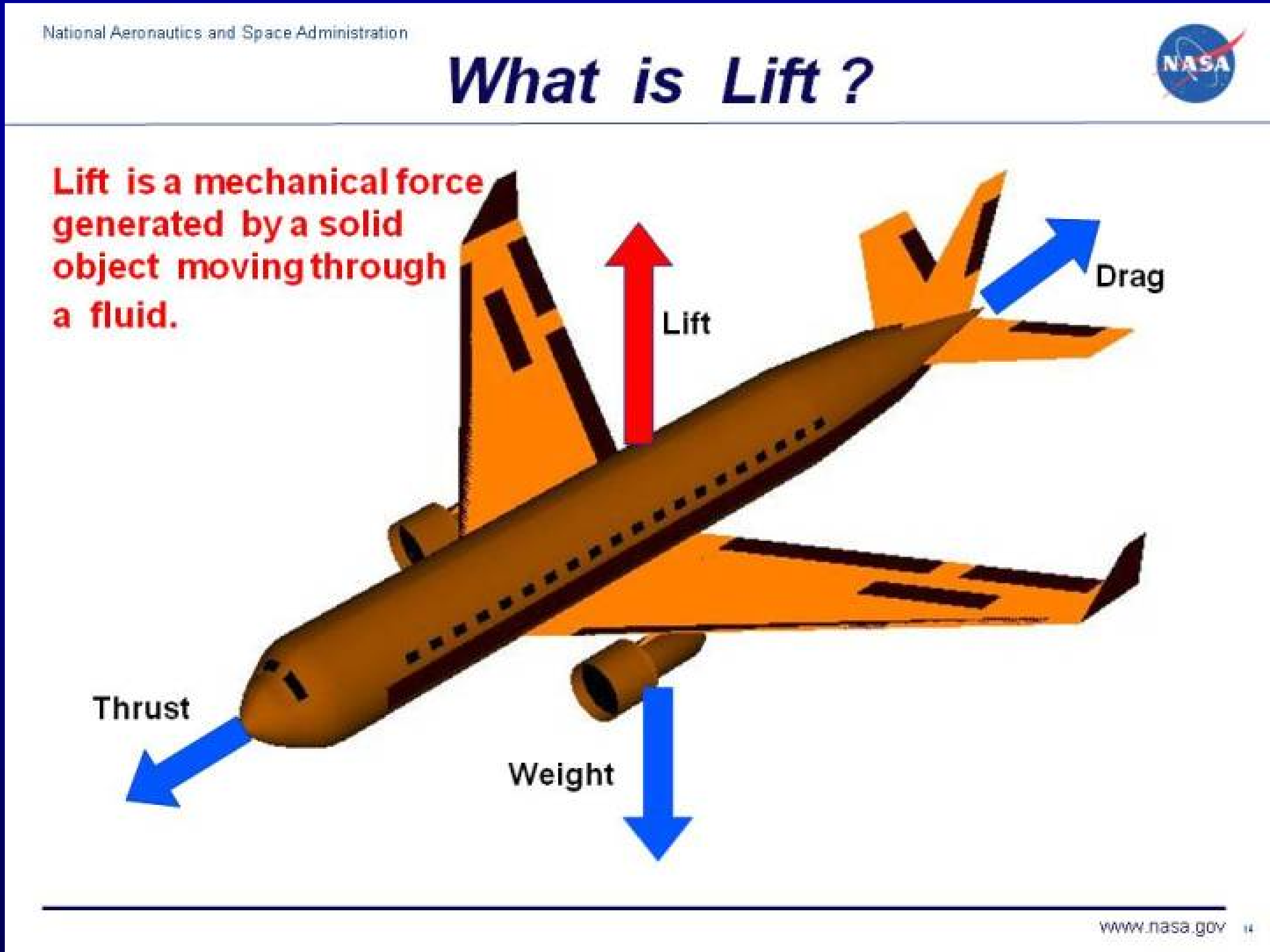



Magnus Force

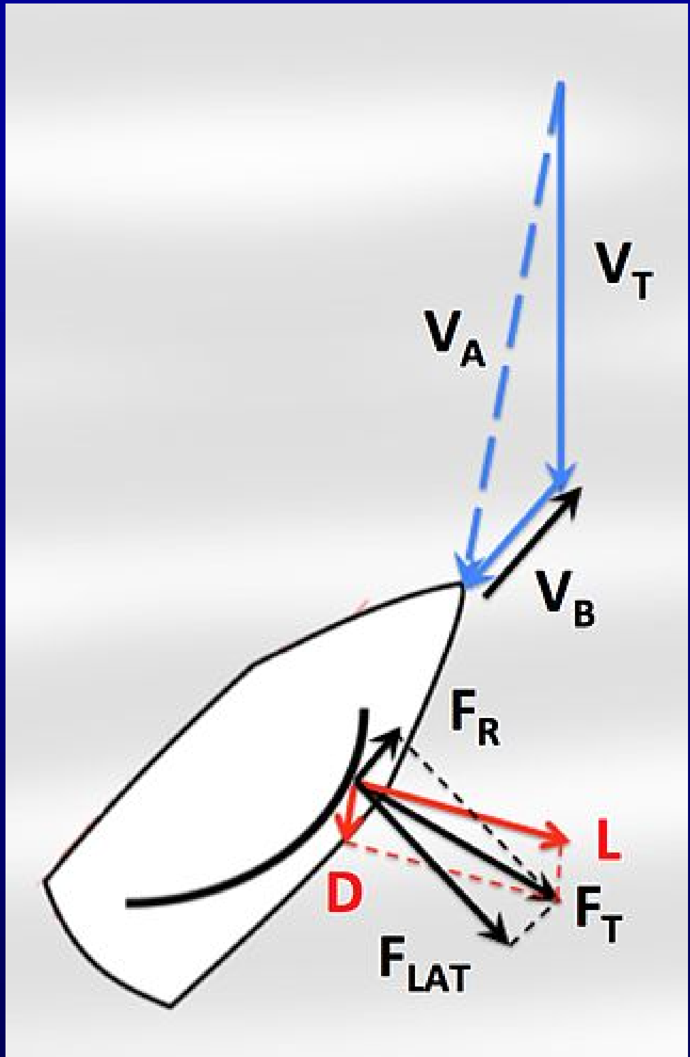
Application: airfoil; or hydrofoil & hovercraft



Assignment:*5.12, 5.13



Lift 升力 **Drag 阻力** **Thrust 推力**



Forces on sails

$\boldsymbol{v}_{\mathrm{T}}$ ----true wind velocity
$\boldsymbol{v}_{\mathrm{B}}$ ----boat velocity
$\boldsymbol{v}_{\mathrm{A}}$ ----apparent wind velocity

Wind forces acting on a sailboat sail (***L*** and ***D***) and being transmitted to the boat ($\boldsymbol{F}_{\mathbf{R}}$—propelling the boat forward—and $\boldsymbol{F}_{\mathbf{LAT}}$—pushing the boat sideways), while close-hauled, are both components of total aerodynamic force ($\boldsymbol{F}_{\mathbf{T}}$).

The rising smoke from a cigarette is at first in laminar flow (smooth column), but it soon becomes the turbulent flow as its speed increases.

The Reynolds number which is defined as

$$\mathrm{Re} = \frac{\rho v L}{\mu}$$

can be used to predict this transition.

*5.5 Symmetries & Conservation laws

Symmetry : physical law is invariant under some operation or transformation.

For example, Newton's laws are invariant under Galilean transformation.

Symmetric transformation is closely related to indistinguishability or unmeasurableness and conservation laws.

Noether's theorem reads:

- **A continuous symmetry property of the Lagrangian density leads to a conservation condition.**

The converse is not true.

We may roughly say:

- **A continuous symmetry property of the system leads to a conservation condition.**

Table 5-1 shows symmetric operation and corresponding conservation.

Conservation of momentum, energy, and angular momentum appear in mechanics;

Charge conservation will be mentioned in § 15.1;

In § 26.5 we will discuss left-right symmetry in biological molecules.

Other part will mainly be discussed in § 30.3 and § 30.5.

Transformation could be continuous or discrete, dependent or independent on space and time.

gauge transform

Different reasoning(in general and theoretical physics):

Usually we discuss conservation laws from equation of motion and then state symmetry.

In theoretical physics, reasoning is different, taking symmetry as starting point.

By 1905 people had realized profound symmetry in Maxwell's equations (§ 20.2).

Einstein thought in 1907, "why couldn't we reverse this procedure? Why couldn't we start from symmetry and derive equations that are consistent with these symmetries and derive experiments which would be in agreement with those equations?".

Yang-Mills' non-Abelian gauge symmetry theory is a model in this aspect. This theory suits electroweak interaction and strong interaction between quarks.

Parallels between Yang-Mills theories and general theory of relativity were first pointed out by Utiyama.

Mathematics Pronunciation Guide

http://www.waukesha.uwc.edu/mat/kkromare/up.html